

Jad Chahin

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EDUCATION

University of Waterloo

Bachelor of Computer Science, Co-op | 89% average

Waterloo, ON

Sep 2025 – Apr 2030

EXPERIENCE

Software Engineer Intern

Generis Group

May 2026 – Aug 2026

Toronto, ON

- Built a Next.js/Prisma reporting platform with employee and reviewer roles for a 30-person B2B events team.
- Integrated Jira, HubSpot, and Google Workspace (Calendar, Tasks, Gmail, Chat), using Gemini to extract work items from emails and chats and to generate daily summaries.
- Developed an LLM-driven system that plans WordPress production changes for human approval, executes them with guardrails and rollback, and verifies rendered pages after each write.
- Constrained the LLM to expressing intent: it never sees record IDs, form keys, or file paths.

Software Engineer Intern

QuickBite Collective

Jun 2025 – Aug 2025

Markham, ON

- Shipped a Toast POS dashboard (Next.js, NestJS, PostgreSQL) for 2 brands and 40 restaurants, replacing manual exports with live reporting.
- Implemented Toast order ingestion with webhooks, BullMQ queues, backfills, and catch-up syncs, processing 41M+ normalized records verified by automated coverage checks.
- Designed idempotent sync jobs with deduplication keys, database lease locks with heartbeat renewal, and cursors that advance only after fully successful runs.

PROJECTS

Derive AI: Math Workspace | React, TypeScript, Express, MongoDB

CXC AI Hackathon 2026

- Built a pressure-aware canvas whiteboard with ink smoothing, selection and group transforms, shape tools, pan/zoom, and undo/redo history.
- Integrated OpenRouter-backed handwriting recognition and math reasoning to solve, graph, check work, and suggest next steps from selected strokes or images.
- Implemented Auth0-protected notes/chat APIs with bulk sync, indexed per-user storage, user-isolation tests, and streaming Gemini responses.

Planet Sim: 3D Physics Engine | C++20, OpenGL, CMake, GLSL

- Implemented a rigid-body engine with velocity-Verlet integration, adaptive substepping, pairwise gravity, swept-broadphase continuous collision detection, friction/restitution, warm-starting, and sleeping.
- Built an instanced renderer with directional shadows, HDR compositing, bloom, fog, and procedural materials.
- Designed focus/orbit camera controls, minimap, path trails, persistent settings, and key rebinding, with tests across physics, rendering, camera, and UI.

Study Planner: Multi-Agent Exam Scheduler | Python, Google ADK, Gemini

- Built a four-agent pipeline that maps PDFs to courses, extracts topics from chunked documents, and estimates per-topic workload.
- Implemented deadline-aware, day-by-day exam scheduling with workload caps, CSV/Markdown export, and feasibility review.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C, C++, Java, SQL

Frontend: React, Next.js, Tailwind CSS, TanStack Query

Backend: Node.js, Express, NestJS, Prisma, PostgreSQL, MongoDB, Redis

AI/LLM: Gemini, OpenRouter, Google ADK, OpenAI API

Testing: Jest, Vitest, Playwright, React Testing Library

Tools: Git/GitHub, GitHub Actions, Docker, Linux/Bash, Vercel, Supabase, Jira